

Smart Shock Protection & Auto-Sensing Outlet

A Low-Voltage Optocoupler & Shunt-Based Zero-Standby Mains Safety System

1. ABSTRACT

Electrical shock accidents in household and industrial environments often occur due to exposed live terminals in power outlets. Traditional wall sockets remain energized at 230V AC regardless of whether an appliance is plugged in or not, posing severe electrocution risks—especially to children or during accidental contact.

This project presents a **Smart Shock Protection & Auto-Sensing Outlet** that keeps the socket completely de-energized from high-voltage mains during standby. The circuit utilizes a low-voltage (5V DC) sensing loop featuring an optocoupler to detect when a physical load is connected. Upon detection, an Arduino triggers a relay to switch the socket to 230V AC mains. While active, the system monitors AC current via a low-value shunt sensing resistor connected to an analog pin. When the load is disconnected, the system instantly cuts the 230V AC supply and reverts the socket back to safe 5V DC standby mode.

2. WHAT IS THE PROBLEM?

Conventional AC power distribution sockets present several fundamental safety hazards:

- **Always-Live Terminals:** Standard wall outlets continuously carry 230V AC potential even when empty or idle.
- **Child & Domestic Hazards:** Foreign conductive objects (e.g., pins, wire fragments, wet fingers) inserted into active outlets cause severe or fatal electrical shocks.
- **Unintentional Contact:** Damaged socket covers, exposed wiring, or accidental contact during maintenance present continuous risk factors.

Key Limitation of Existing Solutions: Standard protection mechanisms like Residual Current Circuit Breakers (RCCBs) or Ground Fault Circuit Interrupters (GFCIs) only trip *after* a fault current flows through a human body or ground loop. They do not prevent the initial high-voltage exposure.

3. THE IDEA

The core concept behind this design is "**Power-on-Demand**" **Zero-Standby Voltage**:

1. Keep 230V AC mains **completely disconnected** from socket terminals when idle.
2. Maintain a safe, non-lethal **5V DC sensing potential** across the socket terminals during standby.
3. Automatically switch to 230V AC **only when a valid load loop is completed** by plugging in an appliance.
4. Continuously monitor load current during operation and **immediately isolate 230V AC** as soon as the load is unplugged or switched off.

4. IDEA IMPLEMENTATION

The system operates in a closed control loop through two main functional states managed by an Arduino microcontroller:

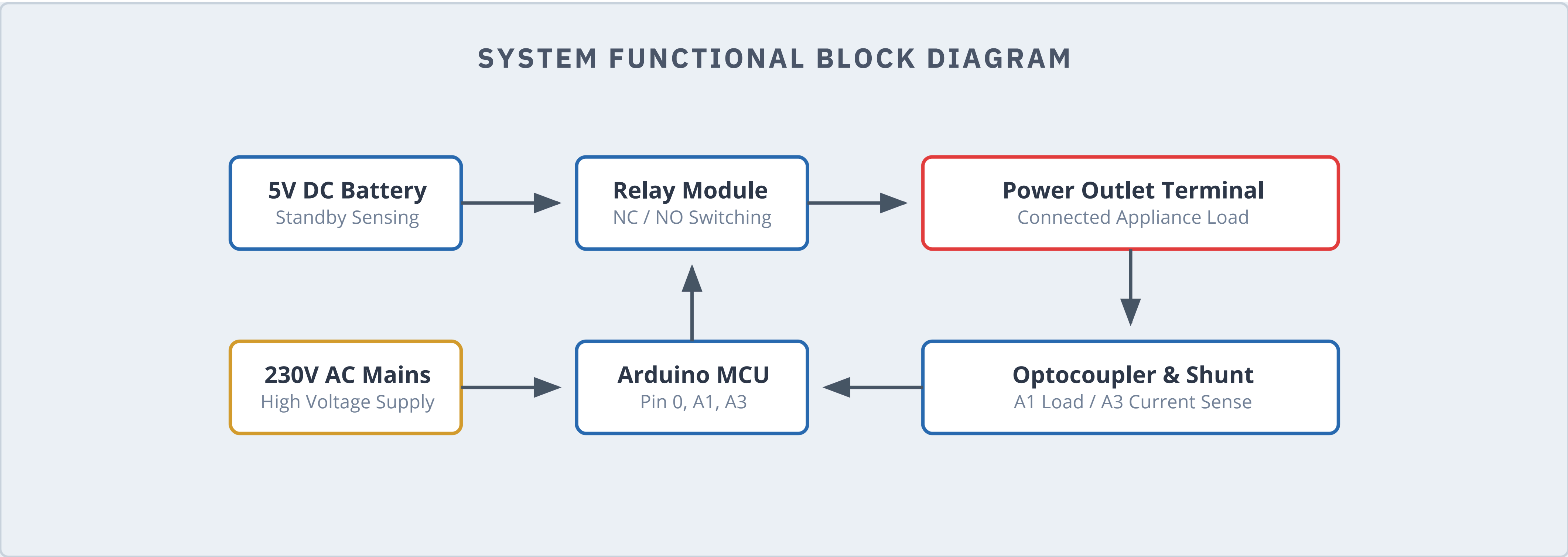
Phase A: Connection Sensing Phase (Idle State)

- The electromechanical relay rests in its **Normally Closed (NC)** position, connecting a 5V DC power source across the socket terminals through an optocoupler input.
- When the socket is empty, the 5V circuit remains open, and the optocoupler’s internal LED remains OFF.
- Plugging in an appliance completes the low-voltage DC loop through the internal resistance of the load.
- Current flows through the optocoupler LED, turning ON its phototransistor and pulling pin **A1** state LOW/HIGH as configured.

Phase B: High-Voltage Energization & Current Monitoring Phase (Active State)

- Recognizing the signal on pin **A1**, the Arduino drives pin **0LOW** to activate the LOW-triggered relay module.
- The relay flips from **NC to NO (Normally Open)**, disconnecting the 5V DC sense line and connecting 230V AC mains directly to the socket load.
- While operating, AC current flows through a low-value shunt resistor connected in series with the load.
- The proportional voltage drop across this resistor is continuously sampled by analog pin **A3**.
- When the appliance is disconnected or switched off, current flow ceases. Pin **A3** detects the drop in voltage and releases the relay (Pin **0 HIGH**), reverting the socket back to safe 5V DC standby mode.

5. CIRCUIT DESCRIPTION & SIGNAL FLOW



Component	Pin / Terminal Connection	Functional Role in Circuit
5V DC Supply	Relay Normally Closed (NC) Terminal	Provides safe low-voltage sensing power during standby mode.

Component	Pin / Terminal Connection	Functional Role in Circuit
230V AC Mains	Relay Normally Open (NO) Terminal	High-voltage supply path engaged only after load verification.
Optocoupler	Relay COM Output & Arduino Pin A1	Isolates high voltage and sends physical connection signal to Arduino.
Relay Module	Arduino Trigger Pin 0 (Active LOW)	Switches power outlet between 5V DC (Sensing) and 230V AC (Operating).
Shunt Resistor	Series with Load & Arduino Pin A3	Monitors AC current drop to trigger automatic shutoff upon removal.

6. APPLICATIONS

The proposed smart socket safety framework can be integrated into several critical domains:

- **Child-Safe Residential Smart Outlets:** Ideal for nursery rooms, homes, schools, and daycare centers where children risk touching energized sockets.
- **Outdoor & Industrial Power Stations:** Prevents accidental shock in wet or high-risk environments (e.g., lawnmowers, outdoor tools, wet construction sites).
- **Laboratory & Educational Benches:** Safeguards students and engineers working in electronics laboratories against open mains contacts.
- **Public Charging Kiosks:** Prevents tampering, vandalism, or accidental electrocution in publicly accessible outlets.

7. CONCLUSION

This project successfully demonstrates a proactive safety mechanism that eliminates standby high-voltage potential on open power sockets. By combining optocoupler-based low-voltage detection with continuous shunt resistor current monitoring, the circuit achieves reliable automatic power delivery only when a genuine load is connected. The system significantly reduces electrocution risks compared to conventional passive socket covers or reactive fault interrupters, offering a robust foundation for next-generation smart outlet safety.



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9. Hardware Images

